Ming Wu

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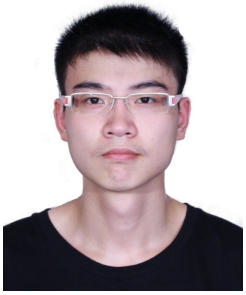

Research Vision

Grounded Causal Intelligence in Manufacturing Research focuses on the synthesis of fundamental manufacturing physics and advanced artificial intelligence to transition from reactive qualification to proactive, "first-time-right" manufacturing. A dual-expertise approach is utilized, combining **Causal AI** with **Adaptive Digital Twins** to resolve the "physical residual", which is the gap between idealized simulations and stochastic reality. The primary objective is the development of a "Symbiotic Co-Learning Loop," where AI agents autonomously design minimal physical experiments to correct physics-based models. Applications focus on zero-shot planning for Additive Manufacturing and intelligent state detection in Subtractive Processes.

Education

Guangdong University of Technology, Mechanical Engineering (ESI TOP 1‰), *Ph.D.* 2016.09 - 2020.12

- **State Key Laboratory**: State Key Laboratory of Precision Electronic Manufacturing Technology and Equipment
- **PhD. Thesis**: Theoretical and experimental research on the surface microfabrication by mask electrolyte jet machining based on multi-physical model and machine learning model
- **Focus**: Integration of multi-physical modeling (thermal, fluid, plasma) with data-driven methodologies for complex subtractive processes.

Guangdong University of Technology, Mechanical Engineering (ESI TOP 1‰), *Master* 2013.09 - 2016.07

- **State Key Laboratory**: State Key Laboratory of Precision Electronic Manufacturing Technology and Equipment
- **Master Thesis**: Modeling and research of mask- Jet-ECM applied in surface micro texture

Hongkong Polytechnic University, Nanomanufacturing lab, *Summer school* 2013.07 - 2013.08

Dongguan City University, Mechanical Engineering and automation, *Bachelor* 2008.09 - 2012.07

Academic Appointments

KU Leuven, Research Associate 2024.03 - present

Joint researcher from *Manufacturing Processes and Systems (MaPS)*, Department of Mechanical Engineering and *Declarative Languages and Artificial Intelligence (DTAI)*, Department of Computer Science

Research field:

- AI-assisted workplan generation for 3D designs with CNC manufacturing processes
- Data-driven process parameter optimization for non-metallic additive manufacturing process
- Application of machine learning and AI technologies for (additive) manufacturing in-process monitoring and control

KU Leuven, Postdoc 2021.03 - 2024.03

Joint postdoc from *Manufacturing Processes and Systems (MaPS)*, Department of Mechanical Engineering and *Declarative Languages and Artificial Intelligence (DTAI)*, Department of Computer Science

Research field:

- Data-driven process parameter optimization for non-metallic additive manufacturing process
- Application of machine learning and AI technologies for (additive) manufacturing in-process monitoring and control
- Computer vision inspection of industrial products

KU Leuven, International scholar 2019.09 - 2020.09

[Award] Scholarship under the Outstanding Young Scientific Research Talents International Cultivation Project Fund of Department of Education of Guangdong Province.

Research field:

- Fabrication of micro-scale surface structures using non-traditional manufacturing process

Tokyo University of Agriculture and Technology, Visiting Scholar 2023.11

- Funding by SAKURA SCIENCE Exchange Program administrated by Japan Science and Technology Agency.

Institute of Science and Innovation in Mechanical Industry Eng., Portugal, Visiting Scholar 2018.01

Urban Transport Systems Laboratory of EPFL, Switerland, Visiting Scholar 2017.07

Research Grants & Secured Funding

- Fields of Reaserch:

- AI-assisted CAD-to-CAM workplan generation and optimization
- Multiphysical Modeling
- In-process Monitoring and Control (based on vision/acoustic data)
- Additive Manufacturing (LPBF/WAAM/FFF) and Subtractive Manufacturing (ECM/EDM)

- Funding

AutoCAM_SBO, Belgium **2024.10 - present**

- **Description** Strategic Basic Research: AI-assisted CAD-to-CAM optimization for Machining Processes
- **Total project budget** ~2.75M Euros
- **Responsibility** Proposal Co-Author & Lead Researcher Work plan generation based on the product specific preference and the environmental constrains by generative AI

MuSIC_SBO, Belgium **2022.10 - present**

- **Description** Strategic Basic Research: Multi Sensor fusion for In-process Control
- **Total project budget** ~3.5M Euros
- **Responsibility** Lead researcher Investigations on AI based defects inspection in Laser Powder Bed Fusion(LPBF) and wire arc additive manufacturing(WAAM) process

C3AF3, Belgium **2022.10 - present**

- **Description** KU Leuven Internal Funding: Adaptive Fused Filament Fabrication
- **Total project budget** ~220K Euros
- **Responsibility** Lead researcher Researches on Data-driven and physical driven process parameter optimization for filament printing process with thermal vision system

CTO Action, Belgium **2022.06 - 2022.12**

- **Description** Flanders make CTO Action 2022
- **Total project budget** ~95K Euros
- **Responsibility** Lead researcher Drafting the glossary for Belgium industrial 5.0 (Triple helix section).

ADAVI, Belgium **2021.03 - 2024.03**

- **Description** Flanders Make Interdisciplinary Collaborative Research project. Anomaly Detection by Active Visual Inspection
- **Responsibility** Researcher 3d measurement by single-eye fringe projection for industrial products.

National Natural Science Foundation of China, China **2016.09-2019.09**

- **Description** A new method and application investigation on micro surface fabrication by mask micro electrochemical discharge machining
- **Responsibility** Lead researcher A novel electrochemical machining process comprising scanning electrode + photolithography was investigated.

National Natural Science Foundation of China, China **2016.09-2019.09**

- **Description** Fundamental studies on precision electrochemical machining of thin-walled micro-grooved plates for difficult-to-machine materials
- **Responsibility** Lead researcher A novel Micro-Electrochemical Discharge Method for machining thin-Walled Structures was studied.

National Natural Science Foundation of China, China **2018.09-2019.09**

- **Description** Investigation on a new approach of manufacturing surface functional microstructures using magnetic powder electrode discharge.
- **Responsibility** Proposal drafter, Lead researcher A novel method of self-adaptive EDM for surface microstructure fabrication was developed.

Grant Proposals Currently Under Review

Horizon Europe (CL4-2025): AI4AM **Submitted Sept 2025**

- **Title** Artificial Intelligence-Driven Qualification for Additive Manufacturing based on Foundation Models.
- **Role:** Proposal Co-Author, Key Investigator, Work Package Contributor (Consortium includes Siemens, Fraunhofer, UCL).
- **Scope:** Developing the first Qualification Foundation Model (QFM) for LPBF, DED, and AFSD processes. Validation across aerospace, medical, and energy sectors.
- **Significance:** Strategic alignment with the ”GenAI4EU” initiative; targets a 50% reduction in qualification time.

Horizon Europe (CL4-2025): SYSTEMATIC **Submitted 2025**

- **Title:** Tools for Life-Cycle Assessment for Safe and Sustainable Design of Chemicals, Materials, and Products.
- **Role:** Key Investigator (Consortium includes Cranfield University, Ansys, British Geological Survey).
- **Scope:** Developing AI-driven Life Cycle Assessment (LCA) tools to support Safe and Sustainable by Design (SSbD) frameworks.

FWO Senior Postdoctoral Fellowship: PRIME **Submitted 2025**

- **Title:** Proactive Reasoning for Intelligent Manufacturing Engineering.
- **Role:** Principal Investigator.
- **Scope:** Development of a ”Causal AI Brain” and ”Adaptive Digital Twin” to enable first-time-right manufacturing.

Global Seed Fund 2026 (China-Flanders): AWARE-SENSEI **Granted 2025**

- **Title:** Construction of Adaptive Wear-Resistant Surfaces and In-Situ Sensing.
- **Role:** Co-Supervisor (Collaboration with Prof. Yi Zhu, Zhejiang University).
- **Scope:** In-process acoustic fingerprinting for Wire Arc Additive Manufacturing (WAAM) to predict microstructural integrity.

- Academic Service

- **300+** review activities for **50+** academic journals.
- **Editor Board** of International Journal of Industrial and Manufacturing Systems Engineering
- Member of The American Society of Mechanical Engineers (ASME) since 2022

- Member of Institute of Electrical and Electronics Engineers (IEEE) since 2022
- Member of Flanders Make since 2021
- Member of Guangdong Mechanical Engineering Society since 2021

PUBLICATIONS

Journal Paper

First/Cooresponding Author

1. **M. Wu**, J. Liu, J. He, X. Chen, and Z. Guo*, Fabrication of surface microstructures by mask electrolyte jet machining, International Journal of Machine Tools and Manufacture 148 no. ARTN 103471, (2020), DOI: 10.1016/j.ijmachtools.2019.103471 IF=18.8
2. **M. Wu**, Z. Yao, P. Karsmakers, M. Verbeke, B. Gorissen, and D. Reynaerts*, Data-driven models with physical interpretability for real-time cavity profile prediction in electrochemical machining processes, Engineering Applications of Artificial Intelligence 160, (2025), 111807, DOI: 10.1016/j.engappai.2025.111807

   IF=8ESI highly cited hot paper
3. J. Ge, Z. Yao, **M. Wu***, J. H. S. Almeida, Y. Jin, and D. Sun, Tackling data scarcity in machine learning-based cfrp drilling performance prediction through a broad learning system with virtual sample generation (bls-vsg), Composites Part B: Engineering, (2025), 112701, DOI: 10.1016/j.compositesb.2025.112701

   IF=14.2ESI hot paper
4. Z. Yao, **M. Wu***, J. Qian, and D. Reynaerts*, Non-invasive radio frequency (rf)-driven in-process monitoring and control for enhancing micro-electrical discharge machining stability and efficiency, Mechanical Systems and Signal Processing 249, (2026), 114072, DOI: 10.1016/j.ymssp.2026.114072

   IF=8.9
5. Z. Yao, **M. Wu***, J. Qian, and D. Reynaerts*, Intelligent discharge state detection in micro-edm process with cost-effective radio frequency (rf) radiation: Integrating machine learning and interpretable ai, Expert Systems with Applications 291, (2025), 128607, DOI: 10.1016/j.eswa.2025.128607
   IF=7.5
6. Z. Yao, J. Ge, P. Cao, **M. Wu***, J. Qian, Y. Li, and D. Reynaerts*, Advancements in process monitoring and quality control for electrical discharge machining: A comprehensive review, Journal of Materials Processing Technology 345, (2025), 119081, DOI: 10.1016/j.jmatprotec.2025.119081 IF=7.5
7. Y. Lin, Y. Lin, H. Wu*, and **M. Wu***, The enhance-fuse-align principle: A new architectural blueprint for robust object detection, with application to x-ray security, Sensors 25 no. 21, (2025), 6603, DOI: 10.3390/s25216603
8. **M. Wu**, M. Hazak Arshad, K. Kumar Saxena, D. Reynaerts, Z. Guo, and J. Liu*, Experimental and numerical investigations on fabrication of surface microstructures using mask electrolyte jet machining and duckbill nozzle, Journal of Manufacturing Science and Engineering -Transactions of the ASME 145 no. 5, (2023), DOI: 10.1115/1.4056570
9. **M. Wu**, Z. Guo, J. Qian, and D. Reynaerts*, Multi-ion-based modelling and experimental investigations on consistent and high-throughput generation of a micro cavity array by mask electrolyte jet machining, Micromachines 13 no. 12, (2022), DOI: 10.3390/mi13122165
10. **M. Wu**, K. Kumar Saxena, Z. Guo, J. Qian, and D. Reynaerts, Fast fabrication of complex surficial micro-features using sequential lithography and jet electrochemical machining, Micromachines 11 no. 10, (2020), DOI: 10.3390/mi11100948
11. Z. Shao#, P. Zhang#, B. Rui, and **M. Wu***, An aero-thermodynamic physics-informed neural network for small-sample performance prediction of variable-speed centrifugal chillers, Energies 19 no. 6, (2026), DOI: 10.3390/en19061563
12. Z. Shao and **M. Wu***, Composite finite-time adrc for flexible-joint manipulators with frequency-domain separation, Processes, (2026), DOI: 10.3390/pr14050863
13. X. Zhang, J. Wang, J. Wei, X. Yuan, and **M. Wu***, A review of non-fully supervised deep learning for medical image

segmentation, Information 16 no. 6, (2025), DOI: 10.3390/info16060433

14. Y. Shi, Y. He, J. Zha*, B. Chen, C. Shi, and **M. Wu***, Design and manufacture of a flexible adaptive fixture for precision grinding of thin-walled bearing rings, Journal of Manufacturing and Materials Processing 9 no. 5, (2025), 139, DOI: 10.3390/jmmp9050139
15. **M. Wu**, Z. Guo, T. Sun, H. Luo, and Q. Song, Investigation on pulse electrocheimical machining by coated electrode, Modern Manufacturing Engineering 2015 no. 11, (2015), 11–14, DOI: 10.3969/j.issn.1671-3133.2015.11.004

Collaboration

16. L. Ye*, C. Guo, **M. Wu**, J. Qian, N. Yu, and D. Reynaerts, Interpretable deep temporal neural networks for in-situ monitoring under varying conditions in micro-electrical discharge machining, Journal of Manufacturing Systems 81, (2025), 222–237, DOI: 10.1016/j.jmsy.2025.05.007 IF=12.3
17. B. Chen, J. Zha*, Z. Cai, and **M. Wu**, Predictive modelling of surface roughness in precision grinding based on hybrid algorithm, CIRP Journal of Manufacturing Science and Technology 59, (2025), 1–17, DOI: 10.1016/j.cirpj.2025.02.004
18. Y. Zhu*, D. Li, C. Zhang, F. Lin, **M. Wu**, and Y. Chen, Abrasive flow machining of laser powder bed fusion fabricated mini-channels: Modelling and verification, Journal of Manufacturing Processes 141, (2025), 36–47, DOI: 10.1016/j.jmapro.2025.02.073
19. H. Ding, Y. Wang, Q. Tang, Y. Sun, **M. Wu**, W. Wang, J. Bai*, H. Xie, H. Yang, C. Zhang, and Y. Zhu*, Enhancing cavitation erosion resistance of 316l stainless steel through simultaneous microstructure modification during l-pbf, Tribology Transactions, (2025), 1–16, DOI: 10.1080/10402004.2025.2476013
20. J. Liang, A. Chen, **M. Wu**, X. Tang, H. Feng, J. Liu*, and G. Xie*, A shellfish-inspired bionic microstructure design for biological implants: Enhancing protection of antibacterial silver-loaded coatings and promoting osseointegration, Journal of the Mechanical Behavior of Biomedical Materials, (2025), 106963, DOI: 10.1016/j.jmbbm.2025.106963
21. D. Li, T. Zhang, **M. Wu**, W. Wang, H. Ding, F. Lin, and Y. Zhu, Cavitation intensity mechanism in hydrodynamic cavitation abrasive finishing, Journal of Manufacturing Science and Engineering -Transactions of the ASME, (2025), 1–17, DOI: 10.1115/1.4068753
22. G. Li, P. Hu, Y. Fan, **M. Wu**, M. Chen, and T. Yu, Anisotropic tensile properties and high-cycle fatigue behavior of ti6al4v fabricated by single and dual laser-powder bed fusion, Materials Science and Engineering: A, (2025), 148640, DOI: 10.1016/j.msea.2025.148640
23. R. Xu, Y. Zhu, J. Wu, P. Huang, **M. Wu**, W. Wang, C. Zhang*, and H. Yang, Hybrid high-temperature wear mechanisms of additive manufactured ti-6al-4v alloy, Tribology International 205, (2025), 110559, DOI: 10.1016/j.triboint.2025.110559
24. J. Ge, J. Zhang, M. Xu, **M. Wu**, Z. Yao, G. Fu, W. Zhang, B. G. Falzon, G. Catalanotti, J. Xu, Y. Jin, and D. Sun, Data-driven cfrp machining performance prediction and optimization: Advances, challenges and future prospects, Thin-Walled Structures, (2025), 113721, DOI: 10.1016/j.tws.2025.113721
25. J. He, W. Chen, J. Wang, **M. Wu**, L. Zhou, R. Chen, and H. Liang, Exploring electrochemical direct writing machining of patterned microstructures on zr702 with polyacrylamide polymer electrolyte, Micromachines 15 no. 9, (2024), DOI: 10.3390/mi15091074
26. Z. Yao, L. Ye, **M. Wu**, J. Qian, and D. Reynaerts, Prediction of crater morphology and its application for enhancing dimensional accuracy in micro-edm, Journal of Intelligent Manufacturing, (2025), DOI: 10.1007/s10845-024-02430-2
27. Z. Yao, J. Shen, **M. Wu**, D. Zhang, and M. Luo, Position-dependent milling process monitoring and surface roughness prediction for complex thin-walled blade component, Mechanical Systems and Signal Processing 198 no. ARTN 110439, (2023), DOI: 10.1016/j.ymssp.2023.110439
28. H. Zhuang, Z. Guo, **M. Wu**, and J. Wang, Experimental study on electrode machining of micro-pit arrays by scanning jet mask, Machinery Design & Manufacture no. 3, (2022), 139–143, DOI: 10.3969/j.issn.1001-3997.2022.03.028
29. M. Bellotti, **M. Wu**, J. Qian, and D. Reynaerts, Tool wear and material removal predictions in micro-edm drilling: Advantages of data-driven approaches, Applied Sciences 10 no. 18, (2020), DOI: 10.3390/app10186357
30. J. Liu, Q. Huang, **M. Wu**, Z. Zou, Z. Lin, Z. Guo, J. He*, and X. Chen*, Electrochemical discharge grinding of metal matrix composites using shaped abrasive tools formed by sintered bronze/diamond, Science And Engineering

Of Composite Materials 27 no. 1, (2020), 346–358, DOI: 10.1515/secm-2020-0038

31. H. Zhuang, Z. Guo, J. Wang, and **M. Wu**, Study of injection molding microstructure of stainless steel based on jet mask electrochemical machining, Journal of Guangdong University of Technology 3, (2020), 70–74, DOI: 10.12052/gdutxb.190101
32. L. Wen, Z. Guo, C. Chen, X. Chen, **M. Wu**, and Y. Deng, The simulation and experimental study of the mask jet-ecm machining inner wall micro-structure, Modern Manufacturing Engineering, (2019), 1–4, DOI: 10.16731/j.cnki.1671-3133.2019.06.001
33. X. Chen, B. Dong, C. Zhang, **M. Wu**, and Z. Guo, Jet electrochemical machining of micro dimples with conductive mask, Journal of Materials Processing Technology 257, (2018), 101–111, DOI: 10.1016/j.jmatprotec.2018.02.035
34. J. He, **M. Wu**, R. Zhang, J. Liu, Y. Deng, and Z. Guo, A one-step hot-embossing process for fabricating a channel with superhydrophobic inner walls, Journal of Manufacturing Processes 36, (2018), 351–359, DOI: 10.1016/j.jmapro.2018.10.036
35. Y. Qin, J. He*, F. Yang, Z. Zhang, Z. Yuan, and **M. Wu**, Extraordinary optical transmission properties of a novel bi-layered plasmonic nanostructure array, Optik 174, (2018), 684–691, DOI: 10.1016/j.ijleo.2018.08.073
36. R. Zhang, Z. Guo, S. Jiang, **M. Wu**, H. Luo, and G. Wang, Preparation of super-hydrophobic surface of sic/2024al composite through chemical etching and its properties, Corrosion Science and Protection Technology 30 no. 1, (2018), 61–66, DOI: 10.11903/1002.6495.2017.068
37. C. Zhou, Y. Zhang, G. Liu, **M. Wu**, and Y. Yao, Experimental study on cathode optimized ecm-shaped hole, Electromachining & Mould, (2016), 20–32, DOI: 10.3969/j.issn.1009-279X.2016.03.006
38. T. Sun, Z. Yu, **M. Wu**, and G. Wang, Development and application of wall surface micro-machining technology, Modern Manufacturing Engineering 2015 no. 12, (2015), 147–152, DOI: 10.3969/j.issn.1671-3133.2015.12.030
39. T. Sun, Z. Yu, **M. Wu**, G. Wang, and L. Cheng, Simulation and experimental study on flow field of the spiral electrode electrochemical micro-machining, Electromachining & Mould, (2015), 30–33, DOI: 10.3969/j.issn.1009-279X.2015.01.008
40. Y. Tang, H. Luo, **M. Wu**, T. Sun, and X. Yin, Developments and applications of jet electrochemical machining, Electromachining & Mould, (2015), 11–15, DOI: 10.3969/j.issn.1009-279X.2015.z1.003

Poster/Conference

First/Cooresponding Author

1. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, Deep learning-based characterization of fused filament fabrication from temporal thermal data, Procedia CIRP 133, (2025), 573–578, DOI: 10.1016/j.procir.2025.02.098
2. **M, Wu**, S. Shukla, B. Vrancken, M. Verbeke, and P. Karsmakers*, Data-driven approach to identify acoustic emission source motion and positioning effects in laser powder bed fusion with frequency analysis, Procedia CIRP 133, (2025), 531–536, DOI: 10.1016/j.procir.2025.02.091
3. **M, Wu**, Z. Yao, R. Abts, P. Karsmakers, M. Verbeke, and D. Reynaerts*, A threshold-free and label-free pipeline for adaptive pulse classification in electrical discharge machining, Procedia CIRP 133, (2025), 692–697, DOI: 10.1016/j.procir.2025.02.118
4. **M, Wu**, Z. Yao, M. Verbeke, P. Karsmakers, and D. Reynearts*, Geometrical feature classification in electrical discharge machining using in-process monitoring and machine learning, Procedia CIRP 137, (2025), 462–467, DOI: 10.1016/j.procir.2025.07.010
5. **M, Wu**, M. Arshad, K. Saxena, J. Qian, and D. Reynaerts*, Profile prediction in ecm using machine learning, Procedia CIRP 113, (2022), 410–416, DOI: 10.1016/j.procir.2022.09.192
6. **M, Wu**, Z. Guo, J. Qian, and D. Reynaerts*, Fabrication of surface micro letters by electrolyte jet mask machining, Procedia CIRP 95, (2020), 827–832, DOI: 10.1016/j.procir.2020.02.261
7. **M, Wu**, Z. Guo*, J. He, and X. Chen, Modeling and simulation of the material removal process in electrolyte jet machining of mass transfer in convection and electric migration, Procedia CIRP 68, (2018), 488–492, DOI: 10.1016/j.procir.2017.12.079
8. **M, Wu**, S. Shukla, B. Vrancken, M. Verbeke, and P. Karsmakers*, *Data-driven approche to identify disortion of acoustic signals by process contextural information in lpbf*, Leuven, Belgium, (2025)
9. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, *Deep learning-based characterization of fused filament fabrication fromtemporal thermal data*, Leuven, Belgium, (2025)
10. **M, Wu**, S. Kushwaha, Q. Jun, D. Reynaerts, and M. Verbeke*, *Linking 3d models and manufacturing operations*

through point-based labelling for cam knowledge digitalization, Leuven, Belgium, (2025)

11. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, *Classification of fused filament fabrication stability from temporal infrared thermal data using x3d*, Leuven, Belgium, (2025)
12. **M, Wu**, J. Qian, M. Verbeke, P. Karsmakers, and D. Reynaerts*, *Surface inspection of highly reflective rotary products using fringe projection with high dynamic range*, Kortrijk, (2024)
13. **M, Wu**, Z. Yao, J. Qian, M. Verbeke, P. Karsmakers, and D. Reynaerts*, *Enhanced pulse classification in electric discharge machining through semi-supervised learning with dynamic time warping*, Kortrijk, (2024)
14. **M, Wu**, S. Kushwaha, J. Qian, D. Reynaerts, M. Verbeke, and P. Karsmakers*, *Representation of machining state and optimization of rough milling efficiency using machine learning*, Antwerp Expo, Belgium, (2023)
15. **M, Wu**, J. Qian, and D. Reynaerts*, *A machine learning model for profile precition in ecm*, Gent, Belgium, (2023)
16. **M, Wu**, Z. Guo*, X. Chen, and C. Zhang, "Investigation on pulse electrochemical machining of aluminum alloy 7075," in *2017 The 8th International Conference on Mechanical and Aerospace Engineering*, Prague, Czech Republic, (2017)
17. **M, Wu**, M. Arshad, K. Saxena, J. Qian, and D. Reynaerts*, "Data-driven framework of ecm: A machine learning model for profile prediction," in *Proceedings of 17th International Symposium on Electrochemical Machining Technology (INSECT 2021)*, Leuven, Belgium (hosted virtually due to Covid-19), KU Leuven - Departement Werktuigkunde, (2021), 231–240, ISBN: 978-9-08289-313-7
18. **M, Wu**, J. Qian, L. Ye, and D. Reynaerts*, "An unsupervised learning method for pulse classification in electrical discharge machining," in *Proceedings of the 23rd international conference of the european society for precision engineering and nanotechnology*, Technical University of Denmark, Copenhagen, Denmark, euspen, (2023), 149–152, ISBN: 978-1-9989991-3-2
19. **M, Wu**, Z. Guo*, Y. Tang, G. Wang, T. Sun, and S. Jiang, "Investigation on pulse electrochemical machining by coated electrode," in *Proceedings of the International Conference of Manufacturing Technology Engineers (ICMTE) 2014*, Jeju KAL, Korea, Korean Society of Production and Manufacturing, (2014), 107–107
20. **M, Wu**, Z. Guo*, and T. Sun, "Developments and applications of jet electrochemical machining," in *Proceedings of the International Conference of Manufacturing Technology Engineers (ICMTE) 2014*, Jeju, Korea, Korean Society of Production and Manufacturing, (2014), 105–105

Collaboration

21. Z. Yao, **M, Wu**, J. Qian, and D. Reynaerts*, Tool wear evaluation and its evolution in micro-edm using radio frequency (rf) signals, Procedia CIRP 137, (2025), 158–163, DOI: 10.1016/j.procir.2025.07.006
22. M. Arshad, **M, Wu**, K. Saxena, J. Qian, S. Huang, and D. Reynaerts*, Experimental investigations into machining characteristics of niobium carbide cermet with µecm / hybrid laser-µecm, Procedia CIRP 113, (2022), 385–391, DOI: 10.1016/j.procir.2022.09.188
23. K. Saxena, M. Arshad, **M, Wu**, J. Qian, and D. Reynaerts*, A 'virtual sensing' framework for µecm/hybrid laser-µecm for monitoring interelectrode gap conditions, Procedia CIRP 113, (2022), 445–452, DOI: 10.1016/j.procir.2022.09.198
24. R. Abts, J. Zhang, **M, Wu**, M. Verbeke, and E. Ferraris*, "Investigation for thermal analysis in comsol for fused filament fabrication," in *Proceedings of the COMSOL Conference 2024 Florence*, (2024)
25. L. Ye, **M, Wu**, K. Saxena, J. Qian, and D. Reynaerts*, "Prediction of milling-muedm tool wear: The benefit of process monitoring and machine learning model," in *Proceedings of the World Congress on Micro and Nano Manufacturing*, Leuven, Belgium, vol. 2022, RESEARCH PUBLISHING, (2022), 263–266, ISBN: 978-981-18-5180-3. DOI: 10.3850/978-981-18-5180-3_RP54-0053
26. M. Arshad, **M, Wu**, K. Saxena, and D. Reynaerts*, "Analysis of passivation during ecm and hybrid laser-ecm through automated current pulse analysis," in *Proceedings of INSECT 2022 conference*, T. Koyano and M. Uchiyama, Eds., Tokyo, Japan, vol. 2022, Tokyo, Japan: The University of Tokyo/ Japan Society of Electrical Machining Engineers (JSEME), (2022), 123–130, ISBN: 978-4-915698-21-7. [Online]. Available: http://www.scoop-japan.com/kaigi/insect/index.html
27. K. Saxena, **M, Wu**, X. Chen, J. Qian, and D. Reynaerts*, "Multiphysics simulation enabled 'virtual sensing' approach for monitoring the parameters in the interelectrode gap during tool-based hybrid laser-electrochemical micromachining," in *Proceedings of 16th International Symposium on Electrochemical Machining Technology (INSECT) 2020*, Fraunhofer IWU, Chemnitz, Germany (held virtually), vol. 2020, VERLAG WISSENSCHAFTLICHE SKRIPTEN, (2020), 95–104, ISBN: 978-3-95735-125-8. [Online]. Available: http://www.verlag-wiss-scripten.de/

Teaching & Mentorship Competencies

Teaching Competencies

- Core Manufacturing & Design: Computer-Aided Manufacturing (CAM), Design Methodologies (CAD/Generative Design), Manufacturing Processes (Subtractive & Additive).
- Applied Engineering Physics: Thermal Process Modeling (Heat transfer in Additive Manufacturing), Applied Fluid Dynamics (Flow fields in ECM/EDM), Machine Tool Kinematics.
- AI & Digitalization: Artificial Intelligence for Manufacturing (Basics to Advanced), Industrial IoT, Cyber-Physical Systems.

Experience & Supervision

- **PhD Supervision** Currently co-supervising **1** PhD candidates Mechanical Engineering within the AutoCAM_SBO project.
- **Master's Thesis Supervision** Daily supervisor for "Co-pilot for function-driven product design" (2024) and assessor for quality prediction in injection molding (2025).
- **Guest Lecture** *AI in Manufacturing* course, KU Leuven (December 2024).
- **Volunteer lecture** of ICARE Guizhou for Online Course Non-Profit program 2021 (May-June)
- **Technical Training** Delivered expert training sessions for NBN (June 2025).

Innovation & Technology Transfer

Software & Open Source Tools

- **SmartECM Web Application:** Developed a real-time ECM processing profile predictor. (Impact: 204 stars / 33 forks on GitHub) .
- **Thermal AI Valorization Portal:** Created a platform for AI in additive manufacturing, demonstrating "Deep Learning-based Characterization of Fused Filament Fabrication", within the framework of C3AF3 project.
- **Fringe Projection Algorithm:** Open-source algorithm for inspecting high-reflectivity surfaces (Impact: 162 stars / 17 forks on GitHub).

Patent

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2. Z. Guo, **M. Wu**, Z. Huang, J. Liu, Y. Zhang, H. Luo, and Y. Wang, *Disc-type scanning electrode mask micro-electrolysis discharge machining system and machining method*, (2020). [Online]. Available: https://patents.google.com/patent/CN105127526B/en?oq=CN105127526B
3. Z. Guo, **M. Wu**, J. Liu, Z. Huang, Y. Zhang, H. Luo, and G. Wang, *Micro-electrolysis discharge machining working solution and micro-electrolysis discharge machining method*, (2020). [Online]. Available: https://patents.google.com/patent/CN105127525B/en?oq=CN105127525B
4. J. He, Z. GUO, **M. Wu**, C. ZHANG, and X. CHEN, *Device and method for locally reinforcing and forming elastic mold for micro-fluidic chips*, (2019). [Online]. Available: https://patents.google.com/patent/US20190039268A1/en?oq=CN106891459B
5. H. Luo, S. Zhan, **M. Wu**, Z. Guo, G. Liu, Y. Zhang, and H. Peng, *A kind of device and electrochemical machining method for the profound and subtle groove of Electrolyzed Processing planar array*, (2019). [Online]. Available: https://patents.google.com/patent/CN106862683B/en?oq=CN106862683B
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8. **M. Wu**, Z. Guo, J. He, and X. Chen, *A kind of device and method of template electro-stream machining metal surface micro-structure*, (2018). [Online]. Available: https://patents.google.com/patent/CN108115233A/en?oq=CN201711385329

9. **M. Wu**, Z. Guo, J. He, and X. Chen, *Micro-electrolysis discharge machining working solution and micro-electrolysis discharge machining method*, (2018). [Online]. Available: https://patents.google.com/patent/CN108149282A/en?oq=CN108149282A
10. X. Chen, Z. Guo, **M. Wu**, L. Wen, X. Zhu, Y. Zhang, and H. Luo, *A kind of device and jet electrochemical machining method for jet Electrolyzed Processing very low power*, (2017). [Online]. Available: https://patents.google.com/patent/CN107052484A/en?oq=CN201710311839
11. Z. Guo, Z. Yao, **M. Wu**, Z. Huang, and J. Liu, *An automatic boosting electrical discharge machining pulse power control method*, (2017). [Online]. Available: https://patents.google.com/patent/CN105195840B/en?oq=CN105195840B
12. Z. Guo, **M. Wu**, S. Jiang, H. Luo, Y. Tang, and T. Sun, *A kind of electrolytic wire turning equipment of the narrow deep trench of revolving body*, (2016). [Online]. Available: https://patents.google.com/patent/CN104174942B/en?oq=CN104174942B
13. H. Luo, **M. Wu**, Z. Guo, S. Jiang, and T. Sun, *Multi-potential electrolytic processing method*, (2016). [Online]. Available: https://patents.google.com/patent/CN104227157B/en?oq=CN104227157B

Miscellaneous

- Languages: English - C1, Mandarin - Native speaker, Dutch - A1
- Administrator of the Channel with ~1k followers